



A real-time semantic 3D vineyard mapping and fruit localization system for yield estimation in dynamic vineyards

Runbin Yang¹, Xilong Zhang¹, Yanyang Chen¹, Xiaotao Huang¹, Sang Feng^{1,*}

School of Electromechanical Engineering, Guangdong University of Technology, Guangzhou, 510006, China

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ABSTRACT

In precision viticulture, autonomous systems are increasingly deployed for inspection and yield estimation. However, mapping 3D vineyard environments and monitoring fruit distribution presents challenges due to fruit occlusion, moving objects, and unstructured backgrounds. To address these issues, this study presents a semantic 3D mapping and fruit localization system designed for yield estimation in dynamic environments. The proposed framework integrates the ORB-SLAM3 backend with a lightweight semantic segmentation model, named Edge Fusion Segmentation Network (EFSegNet). Derived from the PIDNet baseline, EFSegNet introduces a Gradient Feature Fusion Block (GFFM) and a Deformable Spatial Attention (DSA) module. These components are designed to refine fruit boundaries and separate overlapping grape bunches, a requirement for individual counting. The generated semantic masks guide the removal of dynamic feature points to maintain map consistency, while enabling semantic mapping of fruit clusters. Additionally, a three-stage point cloud filtering strategy is applied to refine the global map, followed by DBSCAN clustering and PCA-based modeling to extract the 3D location and quantity of grape bunches. Validation experiments in a real-world dynamic vineyard indicate that the system operated at 23.1 FPS, with an average 3D localization error of 21.02 mm. The average counting error was 5.95%, demonstrating the system's applicability for automated yield estimation.

1. Introduction

In precision agriculture, obtaining accurate spatial data on fruit distribution and yield is fundamental for orchard management [1,2]. Autonomous inspection robots equipped with 3D mapping and yield estimation capabilities assist in optimizing logistics, labor planning, and market forecasting [3,4]. Beyond detection, the 3D localization of fruit clusters supports tasks such as high-throughput phenotyping, maturity monitoring [5], and pest detection [6], providing data for vineyard operations.

Although agricultural robots are utilized in structured settings like greenhouses [7] and standardized orchards [8], perception in unstructured field conditions remains a challenge. Traditional fruit recognition methods, relying on image processing techniques such as color thresholding [9], are affected by variable lighting and cluttered backgrounds in real-world orchards. Deep learning has enhanced feature extraction capabilities. Earlier works adopted object detection networks such as YOLO [10] and R-CNN [11]. To enable volume estimation through contour extraction, research has transitioned towards semantic segmentation networks, including Mask R-CNN [12], DeepLabv3+

[13], BiSeNetv2 [14], SegFormer [15], and PIDNet [16]. Nevertheless, segmenting irregularly shaped, occluded, and touching grape bunches remains a challenge for yield estimation, as noted in recent reviews [17,18]. Lightweight networks often compromise boundary precision for speed, resulting in under-counting of clustered fruits. While Transformer-based architectures [19] offer high accuracy, they impose a computational load that affects real-time deployment on mobile platforms.

To resolve the spatial ambiguity in 2D methods, research has focused on 3D perception [20]. Depth acquisition technologies, including RGB-D cameras, LiDAR, and structured light, have been integrated to facilitate the transition from 2D detection to 3D localization. For instance, instance segmentation combined with geometric analysis has been used for 3D reconstruction of guava fruits [21]. Similarly, algorithms integrating obstacle perception have been proposed for tomato selection [22]. Furthermore, point cloud processing methods have proven instrumental, as exemplified by the combination of Mask R-CNN with PointNet for pomegranate modeling [23]. Slim-Banana integrates a lightweight YOLO-based detector with calibrated TOF depth to recover 3D coordinates from 2D detections [24].

* Corresponding author.

E-mail address: fengsang@gdut.edu.cn (S. Feng).

¹ These authors contributed equally to this work.

A two-stage multimodal method further uses instance masks to regularize depth completion before estimating 3D harvesting points via key-point and pose inference [25]. Despite success in static scenes or offline processing, many systems face difficulties in dynamic real-world orchards populated by moving persons. This necessitates the fusion of perception with Simultaneous Localization and Mapping (SLAM). Approaches such as ORB-Livox [26] have fused LiDAR and visual SLAM for dynamic fruit detection. Semantic mapping solutions using depth cameras have also been explored in citrus orchards [27] and strawberry fields [28]. Monocular SLAM has been used for indoor UAV navigation [29] and tree trait estimation [30]. Multi-sensor fusion strategies [31] and 2D-to-3D conversion mechanisms [32] have also been proposed. However, inspection robots in dynamic vineyards still face issues: (a) map corruption and tracking drift caused by moving persons; (b) computational constraints affecting real-time monitoring; and (c) inaccurate yield counts due to coarse spatial aggregation in dense canopies.

This study addresses these challenges by proposing a real-time semantic 3D mapping and yield estimation system based on the ORB-SLAM3 framework. We introduce a lightweight network, EFSegNet, to generate masks. These masks serve a dual purpose: filtering out dynamic objects to stabilize SLAM tracking, and segmenting grape bunches with defined boundaries to support counting. A spatial clustering strategy leveraging DBSCAN and PCA-based fitting is then applied to determine the 3D position of each grape bunch.

The contributions of this paper are:

1. Development of a 3D mapping and yield estimation framework that integrates semantic perception with ORB-SLAM3, ensuring inspection capabilities in dynamic vineyards with human interference.
2. Proposal of the EFSegNet segmentation model by incorporating a Gradient Feature Fusion Block (GFFM) and Deformable Spatial Attention (DSA) into the PIDNet baseline, introducing a network designed to separate overlapping fruit clusters and handle irregular boundaries for counting accuracy.
3. Implementation of a three-stage point cloud filtering strategy that improves global map quality by sequentially eliminating noise and outliers via voxel grid filtering, statistical outlier removal, and radius filtering.
4. Deployment of a fruit localization and counting pipeline using DBSCAN clustering and PCA-based ellipsoid fitting to extract the 3D coordinates and quantity of grape bunches for yield estimation.

2. Materials and methods

2.1. Hardware configuration

The experimental setup consists of three primary subsystems: the chassis, the sensor suite, and the control unit. The overall architecture is depicted in Fig. 1.

The mobility platform is based on a differential drive chassis measuring $0.776 \text{ m} \times 0.671 \text{ m} \times 0.319 \text{ m}$ ($L \times W \times H$). It weighs 35.6 kg and can carry a payload of up to 45 kg. The platform is capable of speeds up to 1.65 m/s and can navigate slopes up to 25° , with support for manual joystick control. The perception module includes a ZED2 stereo camera and a Robosense Helios 32 LiDAR. The ZED2 functions as the main sensor for real-time fruit detection and 3D mapping. It is an active stereo system with an integrated IMU, offering a depth range of 0.2 to 20 m and a resolution of 2208×1242 pixels at up to 100 FPS. The Robosense Helios 32 LiDAR (360° FOV, $\pm 1 \text{ cm}$ accuracy) is used solely to acquire ground truth point clouds for accuracy validation and does not participate in the real-time mapping loop. A Lenovo Legion Y9000 laptop serves as the central processing unit, featuring an Intel i5-12500H CPU, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU (8 GB VRAM), running Ubuntu 20.04. This unit handles all sensor data processing, semantic

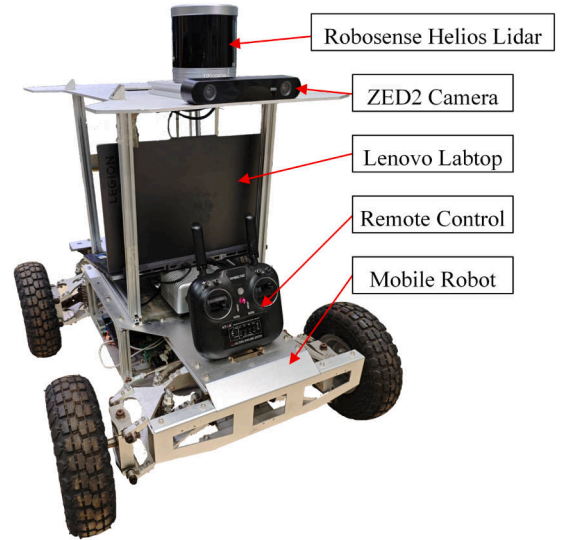


Fig. 1. The experimental robot platform.

segmentation, point cloud mapping, and fruit localization tasks in real-time.

2.2. Data collection and site description

Field experiments were carried out at the Shangguo Pick-Your-Own Orchard in Panyu District, Guangzhou, China ($113^\circ 26' 25.688'' \text{E}$, $22^\circ 54' 41.602'' \text{N}$), as shown in Fig. 2. Situated in the Pearl River Delta, this location offers a warm, humid climate ideal for fruit cultivation. The vineyard represents a typical unstructured agricultural setting, characterized by dense planting, complex intertwined foliage, and fruit occlusion. These conditions provide a realistic testbed for grape detection and localization tasks.

Image sequences of green grapes were captured using the mobile robot as it navigated through the orchard rows. The ZED2 camera was configured to operate at a resolution of 640×360 pixels and a frame rate of 30 FPS. From the recorded sequences, 940 representative RGB images were selected randomly to create the dataset. These images were manually annotated at the pixel level using the LabelMe tool, with pixels categorized into two semantic classes: 'fruit' and 'pedestrian'. This resulting dataset was subsequently used to train and evaluate the semantic segmentation network performance. Sample images from the dataset are presented in Fig. 3.

2.3. System framework

The proposed semantic mapping system, as shown in Fig. 4, integrates four key modules to enable robust 3D mapping and fruit localization: the EFSegNet semantic segmentation module, the ORB-SLAM3 visual backend, the point cloud optimization module, and the fruit clustering/localization module. First, the input stereo images are processed by EFSegNet. The resulting semantic masks are then utilized by the ORB-SLAM3 module to filter out feature points associated with dynamic regions during feature extraction, thereby improving pose estimation stability and map accuracy. Following dynamic point elimination, the system constructs a global 3D point cloud map using depth data from keyframes. This map is semantically enriched by projecting 2D masks onto their corresponding 3D points. To improve map quality, a three-stage filtering process is applied, utilizing voxel downsampling, statistical outlier removal, and radius filtering to remove noise and redundant points. Finally, the refined fruit point cloud is clustered using the Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

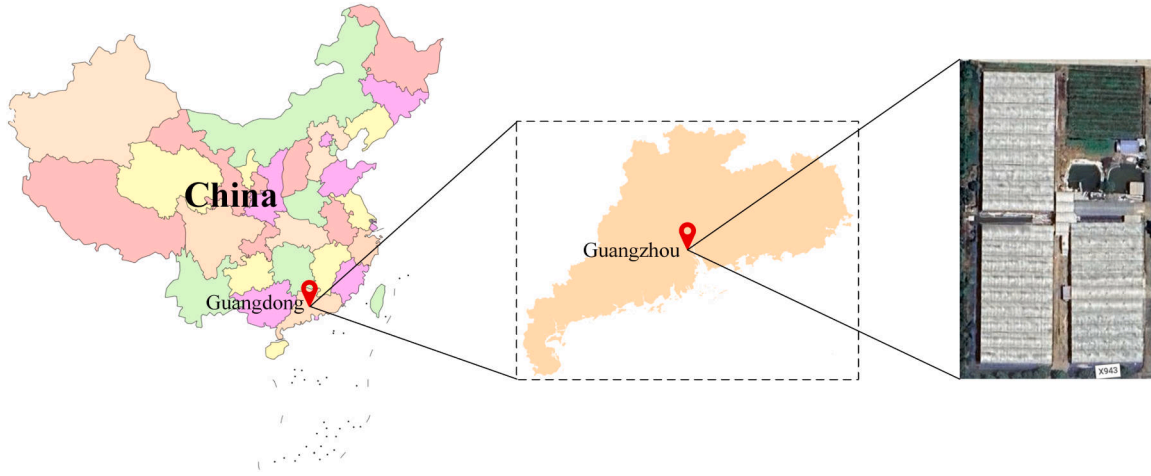


Fig. 2. The vineyard experimental environment.



Fig. 3. Sample images from the vineyard dataset. (a) Static grape bunches. (b) A dynamic scene containing a pedestrian.

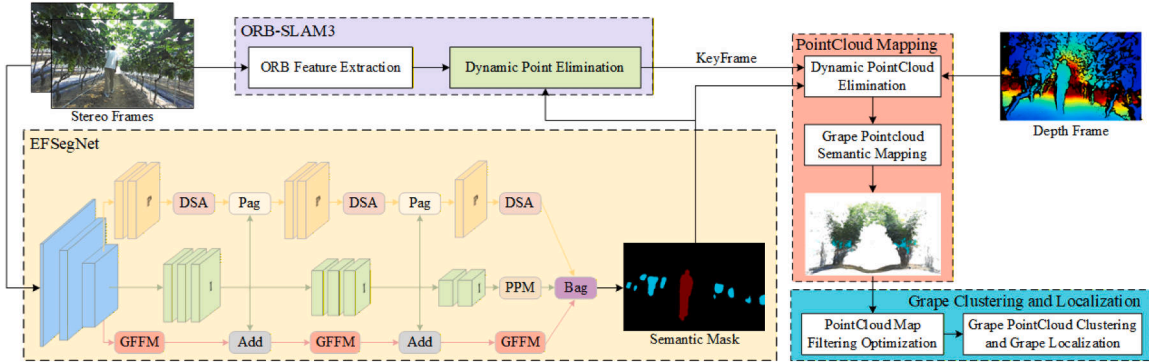


Fig. 4. Architectural overview of the proposed system. It integrates the EFSegNet segmentation module, ORB-SLAM3 backend, point cloud optimization, and fruit localization modules.

algorithm. Principal Component Analysis (PCA) is then applied to model each cluster as an ellipsoid, allowing for the extraction of 3D positions and the counting of grape bunches.

2.3.1. Segmentation network

Reliable yield estimation depends on the ability to distinguish individual grape bunches, particularly when they are touching or overlapping. Standard real-time networks often struggle to define the subtle boundaries between adjacent clusters, resulting in under-counting. To solve this, we introduce EFSegNet, a real-time semantic segmentation network derived from the PIDNet backbone, specifically tuned for vineyard inspection.

As illustrated in Fig. 5, EFSegNet maintains a three-branch structure: spatial, context, and edge. To enhance edge feature extraction, we replace the standard convolutional modules in the edge branch with

a Gradient Feature Fusion Block (GFFM). This module facilitates the fusion of spatial and semantic information by providing explicit edge guidance. Additionally, a Deformable Spatial Attention (DSA) module is integrated into the spatial branch to improve the network's sensitivity to anisotropic targets, such as irregularly shaped grape clusters.

While PIDNet includes an edge-guided design, its original standard convolutions are insufficient for capturing the complex edge gradients found in vineyards. Our proposed GFFM addresses this by integrating three types of differential convolutions to heighten sensitivity to edge gradient changes, as shown in Fig. 6. A lightweight residual path with a gated attention mechanism is also added to compensate for feature degradation.

The central difference convolution (CDConv) [33] highlights local edge variations by subtracting the central response from the output of a

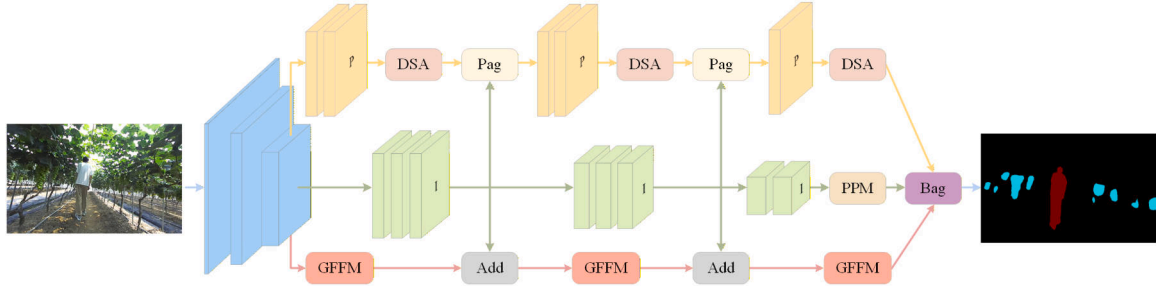


Fig. 5. The framework of EFSegNet. It features a spatial branch with Deformable Spatial Attention (DSA), a context branch, and an edge branch enhanced with the Gradient Feature Fusion Block (GFFM).

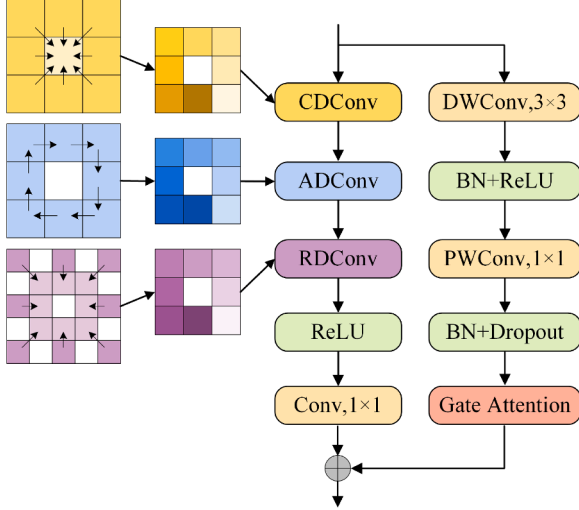


Fig. 6. Gradient Feature Fusion Block (GFFM). This module integrates differential convolutions to enhance edge sensitivity.

conventional convolution. First, the output of the standard convolution is computed as:

$$f'(x) = \text{Conv}(x, W) \quad (1)$$

Where $f'(x)$ is the output of standard convolution. Next, we define the central response term W_{center} by summing all the kernel weights $W_{(i,j)}$ of the convolution kernel W across its spatial dimensions:

$$W_{\text{center}} = \sum_{i,j} W_{(i,j)} \quad (2)$$

We introduce a modulation factor $\theta \in [0, 1]$ to control the proportion of the central difference term. The output of the central difference convolution $f_{\text{cd}}(x)$ is obtained by:

$$f_{\text{cd}}(x) = f'(x) - \theta \cdot \text{Conv}(x, W_{\text{center}}) \quad (3)$$

Subsequently, the angular difference convolution (ADConv) simulates directional variations. $\text{Rotate}(W)$ represents a predefined angular reordering of the kernel weights. The rotated kernel W_{rot} is obtained by subtracting the rotated kernel scaled by θ from the original kernel. The angular difference convolution is formulated as:

$$\text{Rotate}(W) = [W_3, W_0, W_1, W_6, W_4, W_2, W_7, W_8, W_5] \quad (4)$$

$$W_{\text{rot}} = W - \theta \cdot \text{Rotate}(W) \quad (5)$$

$$f_{\text{ad}}(x) = \text{Conv}(x, W_{\text{rot}}) \quad (6)$$

Finally, the radial difference convolution (RDConv) is constructed using a 5×5 kernel with a radially symmetric weighting scheme. The

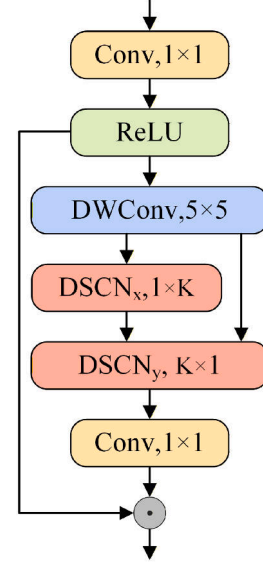


Fig. 7. Deformable Spatial Attention (DSA) module. It uses single-axis deformable convolution to adapt to irregular object geometries.

kernel W_{rd} is defined as:

$$W_{\text{rd}}[i, j] = \begin{cases} +W_k, & (i, j) \in \text{forward radial positions} \\ -\theta \cdot W_k, & (i, j) \in \text{backward radial positions} \\ (1 - \theta) \cdot W_0, & (i, j) = \text{central position} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

The output $f_{\text{rd}}(x)$ is obtained by convolving input x with W_{rd} . The feature is then activated by ReLU and compressed using a 1×1 convolution. The final output of the GFFM (f_{GFFM}) is the element-wise sum of the differential path output and the residual path output f_{res} , as formulated in Eqs. (8) and (9):

$$f_{\text{diff}} = \text{Conv}_{1 \times 1}(\text{ReLU}(x)) \quad (8)$$

$$f_{\text{GFFM}} = f_{\text{diff}} + f_{\text{res}} \quad (9)$$

The DSA module [34], as shown in Fig. 7, is integrated into the spatial branch to handle anisotropic features. It processes input features through 1×1 convolution and ReLU, followed by single-axis deformable convolutions (DSCN) along horizontal and vertical directions, as formulated in Eqs. (10)–(13). Finally, a spatial attention map is generated and applied to the features, as formulated in Eqs. (14) and (15):

$$x' = \text{ReLU}(\text{Conv}_{1 \times 1}(x)) \quad (10)$$

$$x_c = \text{DWConv}_{5 \times 5}(x') \quad (11)$$

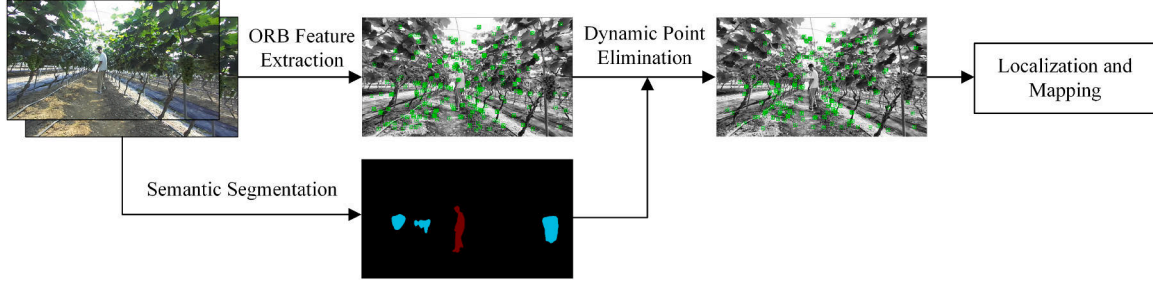


Fig. 8. Dynamic feature point removal process.

$$x_{\text{attn-x}} = \text{DSCN}_x(x_c) \quad (12)$$

$$x_{\text{attn-y}} = \text{DSCN}_y(x_{\text{attn-x}}, x_c) \quad (13)$$

$$\text{atten} = \text{Conv}_{1 \times 1}(x_{\text{attn-y}}) \quad (14)$$

$$x_{\text{output}} = x' \odot \text{atten} \quad (15)$$

2.3.2. Dynamic feature point removal

To eliminate dynamic feature points in dynamic environments and enhance the robustness of the ORB-SLAM3 system in dynamic scenes, a semantic segmentation module is integrated into the original ORB-SLAM3 framework. Specifically, an additional semantic segmentation thread is introduced. In this thread, each RGB frame is processed by EF-SegNet, to produce a corresponding semantic mask. The semantic mask is subsequently synchronized and forwarded to the front-end tracking thread. As illustrated in Fig. 8, during the frame initialization stage, the system first extracts ORB feature points from the input image. Subsequently, it obtains the spatial coordinates of these feature points and queries the corresponding semantic labels from the semantic mask. If a label indicates a dynamic object, the associated feature point is removed; otherwise, it is retained. Therefore, by leveraging the semantic mask, all dynamic feature points within the regions of moving objects can be eliminated, while only the static feature points from the background are preserved for subsequent tracking and mapping. This process reduces matching errors and trajectory drift caused by dynamic object motion, thereby improving the robustness of the SLAM system in dynamic environments.

2.3.3. Semantic mapping and map optimization

During dense point cloud reconstruction, grape fruits are semantically annotated within the point cloud to enhance the semantic representation of the map. Semantic labels are assigned to grape-related points based on the segmentation results. For visualization purposes, a specific color C_{grape} , which is set to cyan in this study, is assigned to the grape point cloud. This facilitates subsequent tasks such as classification, recognition, and human-robot interaction. The rules for point cloud color assignment are defined as Eq. (16):

$$C_i = \begin{cases} C_{\text{grape}} = [0, 255, 255], & \text{label}_i = \text{'grape'} \\ \text{discard}, & \text{label}_i = \text{'pedestrian'} \\ C_{\text{RGB}}(u_i, v_i), & \text{label}_i = \text{'other'} \end{cases} \quad (16)$$

Where $C_{\text{RGB}}(u_i, v_i)$ represents the pixel color in the RGB image corresponding to the i -th point in the point cloud, label_i indicates the semantic label of the i -th point, C_i denotes the assigned color of the i -th point in the point cloud. Due to dense plant spacing, complex branch and leaf structures, and external disturbances such as natural illumination and wind, the reconstructed dense map may present local density anomalies, outliers in the point cloud, and isolated points near plant boundaries and supporting frames. Voxel Grid Filtering is a widely adopted

point cloud downsampling method. It divides the 3D space into equally sized voxel grids and replaces all points within each voxel with their centroid. This process reduces point redundancy and improves computational efficiency while preserving the overall structural integrity of the scene. For each voxel V_k , the representative point p_{voxel} is computed as the mean position of all points $p_i \in V_k$, as defined in:

$$p_{\text{voxel}} = \frac{1}{N} \sum_{i=1}^N p_i, \quad \forall p_i \in V_k \quad (17)$$

To mitigate local density anomalies in the point cloud, we adopt the statistical outlier removal method. Statistical outlier removal identifies outlier points by statistically analyzing the local neighborhood of each point and detecting inconsistencies in point density. Based on local statistical characteristics, abnormal points are removed from the point cloud map. For each point p_i , the average distance μ_i to its k nearest neighbors is computed as in Eq. (18):

$$\mu_i = \frac{1}{k} \sum_{j=1}^k \|p_i - p_j\| \quad (18)$$

Outliers are further removed based on global statistical characteristics. Specifically, a point is considered an outlier and removed if its average distance exceeds the global mean $\bar{\mu}$ plus σ times the standard deviation as in Eq. (19). Meanwhile, a tolerance parameter λ is introduced to adjust the strength of the statistical outlier removal process:

$$\mu_i > \bar{\mu} + \lambda \cdot \sigma \quad (19)$$

To further eliminate spatially isolated points, we introduce the radius outlier removal method. This approach evaluates the number of neighboring points within a fixed radius r for each point. If the number of neighbors $N_r(p_i)$ is less than the minimum threshold $N_{\min}$, the point is considered an outlier and removed. The filtering criterion is described in Eq. (20):

$$N_r(p_i) < N_{\min} \quad (20)$$

2.3.4. Fruit clustering and localization

To achieve spatial localization of fruit targets in point clouds, we propose a clustering method based on DBSCAN (Density-Based Spatial Clustering of Applications with Noise). This method utilizes semantic segmentation masks to identify and isolate spatially independent grape regions within the point cloud. By clustering grape points according to their spatial density distribution, the algorithm separates distinct grape clusters corresponding to different grape bunches. For any 3D point $p_i = (x_i, y_i, z_i)^T \in \mathbb{R}^3$, we calculate the number of neighboring points located within a Euclidean distance ϵ , as defined by $N_\epsilon(p_i)$. If $|N_\epsilon(p_i)| \geq \text{MinPts}$, then p_i is classified as a core point, and its neighboring points are assigned to the same cluster. Clusters are formed through neighborhood connectivity between core points, while points that do not belong to any cluster are considered noise, as defined in:

$$N_\epsilon(p_i) = \left\{ p_j = (x_j, y_j, z_j)^T \in \mathbb{R}^3 \mid \|p_j - p_i\|_2 \leq \epsilon \right\} \quad (21)$$

To accommodate the spatial scale of grape fruits, the neighborhood radius ϵ is set to 0.05m, and the minimum number of points MinPts is set to 50, to avoid misclassifying isolated points as fruit targets. Additionally, the maximum number of points in a single cluster is limited to 5000 to prevent the merging of occluded fragments. After completing spatial clustering, PCA is applied to each grape fruit cluster to extract its geometric structure and perform ellipsoidal modeling. For the k -th cluster, denoted as $C_k = \{p_1, \dots, p_n\}$, the geometric center of the point cloud is first computed and used as the spatial position of the fruit cluster in the global coordinate system, as defined in:

$$c_k = \frac{1}{n} \sum_{i=1}^n p_i \quad (22)$$

The point cloud is first zero-centered, and the covariance matrix is then constructed as in Eq. (23):

$$\Sigma_k = \frac{1}{n} \sum_{i=1}^n (p_i - c_k)(p_i - c_k)^T \quad (23)$$

By performing eigenvalue decomposition on the covariance matrix Σ_k , a direction matrix $R_k \in \mathbb{R}^{3 \times 3}$ is obtained. The column vectors of R_k define the principal axes of the point cloud. The clustered points are then projected onto this principal axis coordinate system. The maximum extent along each axis (a_k, b_k, c_k) is used to compute the semi-axis lengths of the ellipsoid, as defined in Eq. (24):

$$a_k = \frac{1}{2}(x'_{\max} - x'_{\min})b_k = \frac{1}{2}(y'_{\max} - y'_{\min})c_k = \frac{1}{2}(z'_{\max} - z'_{\min}) \quad (24)$$

Where x' , y' , and z' represent the coordinate components of the clustered points in the principal axis coordinate system. Each cluster is ultimately modeled as a 3D ellipsoid with center c_k and semi-axis lengths (a_k, b_k, c_k). The estimated volume of the ellipsoid is given by Eq. (25):

$$V_k = \frac{4}{3} \pi a_k b_k c_k \quad (25)$$

To ensure the physical plausibility of the clustering results, a volume-based filtering range is applied. Only clusters satisfying $V_k \in [V_{\min}, V_{\max}]$ are retained, where $V_{\min} = 0.001 \text{ m}^3$ and $V_{\max} = 0.05 \text{ m}^3$.

3. Experiments and results

To quantitatively evaluate the contribution of each proposed module, we conducted a series of experiments focusing on six key aspects: semantic segmentation performance analysis (Section 3.1), dynamic elimination performance analysis (Section 3.2), 3D reconstruction in a dynamic vineyard (Section 3.3), 3D point cloud map optimization (Section 3.4), fruit localization and bunch counting (Section 3.5), and overall system real-time performance (Section 3.6).

3.1. Semantic segmentation performance

3.1.1. Semantic segmentation comparative experiments

The EFSegNet semantic segmentation network was trained on a system with an Intel i5-12500H CPU and an NVIDIA RTX 3060 GPU, running the Ubuntu 20.04 operating system. The deep learning framework used was PyTorch 1.11. The dataset was partitioned into training, validation, and test sets at an 8:1:1 ratio. The network was trained using the configuration parameters detailed in Table 1.

To provide a comprehensive evaluation of the proposed EFSegNet network's semantic segmentation performance, we conducted a systematic comparison against several state-of-the-art networks: DeepLabv3+, BiSeNetv2, SegFormer-B0, DDRNet-23-slim [35], and PIDNet. All experiments were performed on the vineyard dataset. To ensure a fair and reproducible comparison, all networks were trained for 60,000 iterations with an initial learning rate of 1e-2, using the SGD optimizer. We employed six standard evaluation metrics to assess network performance: mean Intersection over Union (mIoU), mean Accuracy (mAcc), Frames Per Second (FPS), Flops(G), Params(M), and Model Size(M). Here, mIoU

Table 1

Configuration parameters for EFSegNet training.

Parameter Type	Parameter Setting
Optimizer	SGD
Learning Rate	0.01
Momentum	0.9
Weight Decay	0.005
Batch Size	12
Iterations	60,000

represents the average of the IoU values across all classes and is the standard metric for overall segmentation precision. mAcc is the average recognition accuracy for each class, which is particularly useful for evaluating performance on small targets or in scenarios with class imbalance. Flops(G) and Params(M) measure the computational complexity of the network, with Flops(G) representing the number of floating-point operations per second and Params(M) denoting the number of trainable parameters in millions. Model Size(M) quantifies the network's memory footprint. FPS measures the network's operational speed. The metrics are calculated based on the following: TP_i , FP_i , and FN_i represent the number of true positives, false positives, and false negatives for class i , respectively, and k is the total number of classes (two in this study: 'fruit' and 'pedestrian'). The relevant formulas are as follows Eqs. (26)–(28):

$$\text{mIoU} = \frac{1}{k+1} \sum_{i=0}^k \frac{TP_i}{TP_i + FP_i + FN_i} \quad (26)$$

$$\text{mAcc} = \frac{1}{k+1} \sum_{i=0}^k \frac{TP_i}{TP_i + FN_i} \quad (27)$$

$$\text{FPS} = \frac{\text{FrameNums}}{\text{ElapsedTime}} \quad (28)$$

The results of the semantic segmentation experiments are presented in Table 2. On the vineyard dataset, EFSegNet achieved a mIoU of 89.45% and a mAcc of 93.55% with an inference speed of 120.92 FPS. Compared to PIDNet, EFSegNet improved by 1.01% in mIoU and 0.65% in mAcc, attaining the best segmentation performance among all evaluated models. With the integration of the GFFM and DSA modules, both the FLOPs and the number of parameters slightly increased relative to PIDNet, but the increase remains within an acceptable range. Although the inference speed of EFSegNet is lower than that of DDRNet, it remains sufficient for the real-time requirements of the proposed system while delivering superior segmentation accuracy. Therefore, EFSegNet achieves a balanced trade-off between accuracy and efficiency, maintaining high segmentation precision while satisfying the real-time performance demands of the system.

To complement the quantitative metrics and provide an intuitive validation of EFSegNet's superior performance, a qualitative analysis of the segmentation outputs from the test set is presented. These visualizations are chosen to demonstrate the models' effectiveness in handling the key challenges of irregular object boundaries, severe occlusion, and dynamic scene elements. Fig. 9 illustrates the models' segmentation performance under conditions of irregular grape bunch contours and severe occlusion. While methods such as DeepLabv3+, BiSeNetv2, and DDRNet-23-slim can approximate the location of the fruit regions, they generate coarse masks with imprecise boundaries. Segformer-B0 fails to detect the occluded target areas entirely. PIDNet approximately captures the occluded targets, but its segmentation quality remains poor. This deficiency arises because the baseline PIDNet's edge branch employs standard convolutions, which struggle to capture complex background gradient variations. Consequently, it fails to achieve complete segmentation of occluded targets. In contrast, the GFFM module within EFSegNet enhances sensitivity to edge gradient changes by integrating three differential convolutions, enabling finer contour extraction. Therefore, EFSegNet exhibits superior segmentation performance in this challenging

Table 2
Comparison results of classic semantic segmentation networks.

Segmentation Method	mIoU(%)	mAcc(%)	Flops(G)	Params(M)	FPS	Model Size(M)
DeepLabv3 +	87.85	92.85	158.72	41.217	27.4	357.9
BiSeNetv2	87.60	89.93	10.824	3.342	273.04	128.7
Segformer-B0	86.24	96.46	6.857	3.715	177.7	37.7
DDRNet-23-slim	88.02	92.88	4.065	5.732	321.94	46.4
PIDNet	88.41	92.9	19.854	28.759	162.16	118.4
EFSegNet	89.45	93.55	21.786	30.779	120.92	124.6

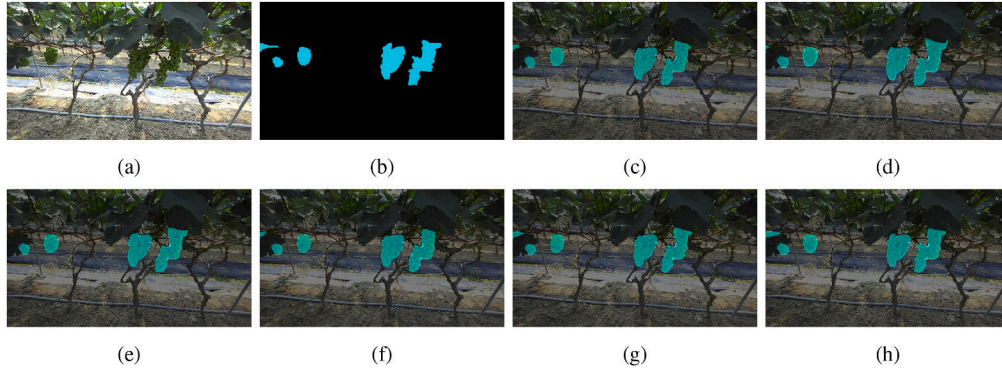


Fig. 9. Comparison of segmentation results on the test set. (a) Image. (b) Ground truth (c) DeepLabv3 + . (d) BiSeNetv2. (e) Segformer-B0. (f) DDRNet-23-slim. (g) PIDNet. (h) EFSegNet.

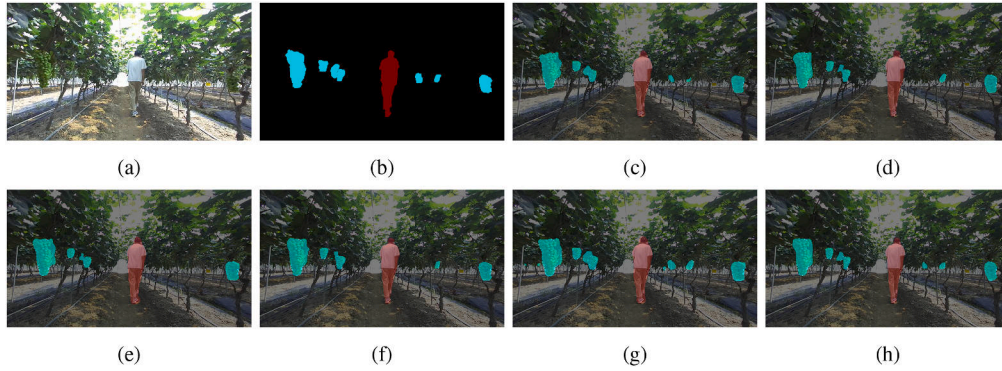


Fig. 10. Comparison of segmentation results on the test set. (a) Image. (b) Ground truth. (c) DeepLabv3 + . (d) BiSeNetv2. (e) Segformer-B0. (f) DDRNet-23-slim. (g) PIDNet. (h) EFSegNet.

static scenario. [Fig. 10](#) demonstrates the segmentation performance in a dynamic scene, a critical test for SLAM robustness. Here, BiSeNetv2, Segformer-B0, and DDRNet-23-slim all fail to identify the distant grape bunch targets. DeepLabv3+ provides incomplete segmentation of the distant targets, while PIDNet produces erroneous segmentation. EFSegNet again outperforms the other models, a result attributed to the synergistic effects of two key components: first, the DSA module provides an adaptive receptive field, enabling better perception of distant grape targets; and second, the GFFM module provides more precise, lower-noise edge priors for both the dynamic pedestrians and the static grape bunches. This, in turn, guides the spatial branch and the semantic segmentation fusion, enabling the generation of a more complete mask.

3.1.2. Ablation study

To further evaluate the effectiveness of each proposed module, a series of ablation experiments were conducted under consistent training and testing conditions, with the results summarized in [Table 3](#). Using PIDNet as the baseline, we systematically analyzed the individual contributions of the GFFM module, its residual path, and the DSA module to overall network performance. Besides mIoU, we introduced the Boundary F-measure to specifically evaluate the quality of boundary

segmentation, as mIoU is dominated by internal object pixels and may not fully reflect edge improvements. Integrating the GFFM module into the edge branch of PIDNet enhanced the network's ability to extract edge features, improving the mIoU to 88.92%. Notably, the Boundary F-measure increased from 78.32% to 80.28%, demonstrating that GFFM effectively sharpens fruit boundaries. The addition of a residual path within the GFFM further mitigated gradient attenuation and stabilized feature propagation, resulting in an mIoU increase to 89.12% and a Boundary F-measure increase to 81.92%. When both the GFFM and DSA modules were incorporated, EFSegNet achieved a well-balanced trade-off between global context modeling and local geometric perception. This configuration yielded the best performance, with an mIoU of 89.45% and a Boundary F-measure of 82.78%.

Compared with the original PIDNet, EFSegNet demonstrated significant improvements in segmentation accuracy with only a marginal and acceptable reduction in inference speed. These results confirm the effectiveness of the proposed modules in enhancing edge feature extraction and optimizing multi-scale feature representation. To qualitatively validate the efficacy of the proposed modules at the feature-representation level, as shown in [Figs. 11](#) and [12](#), we provide a visual comparison of the activations within the edge branch and the bag layer for both EF

Table 3
Ablation study results of the GFFM and DSA modules.

Method	GFFM	GFFM Residual Path	DSA	mIoU(%)	Boundary F-measure(%)	mAcc(%)	FPS
PIDNet	×	×	×	88.41	78.32	92.90	162.16
EFSegNet	✓	×	×	88.92	80.28	93.20	143.78
EFSegNet	✓	✓	×	89.12	81.92	93.37	135.46
EFSegNet	✓	✓	✓	89.45	82.78	93.55	120.92

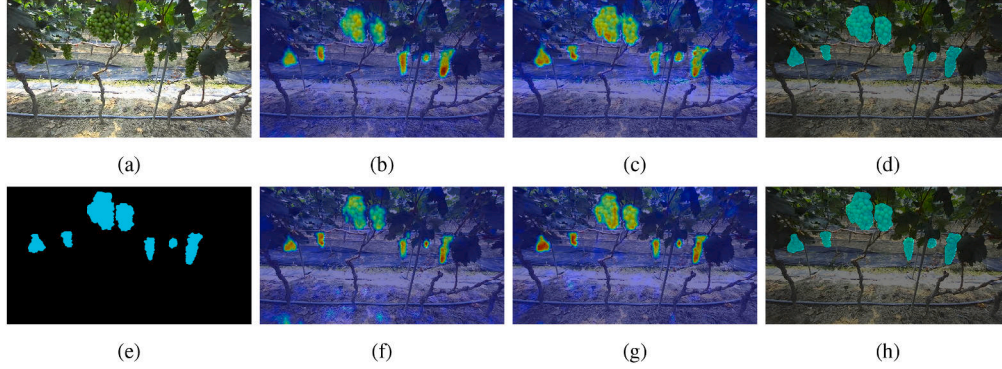


Fig. 11. Comparison of segmentation results between PIDNet and the proposed EFSegNet in the horizontal sequence. (a) Image. (b) PIDNet edge branch output. (c) PIDNet Bag layer output. (d) PIDNet segmentation result. (e) Ground truth. (f) EFSegNet edge branch output. (g) EFSegNet Bag layer output. (h) EFSegNet segmentation result.

SegNet and the baseline PIDNet. The activation heatmaps from the baseline PIDNet's edge branch are diffuse, fail to capture grape bunch contours accurately, and exhibit noise artifacts. Conversely, the EFSegNet edge branch, integrated with GFFM, generates significantly enhanced activation regions. Its hotspots are more concentrated, sharper, and align closely with the actual grape cluster edges. This precise edge prior from the GFFM subsequently propagates to the Bag layer, which, in turn, makes EFSegNet's feature response for the main fruit body demonstrably more focused and confident. This visually confirms that the GFFM module, through its integration of differential convolutions, significantly enhances the network's perception of complex boundaries and fine-grained structures. In scenarios involving dynamic pedestrians and significant target scale variations, the baseline PIDNet reveals its limitations. Its Bag layer almost completely fails to activate on small-scale, distant grape targets, demonstrating a limited receptive field. In contrast, EFSegNet successfully overcomes this challenge. This success is attributed to the synergistic effect of the GFFM and DSA modules. First, the GFFM enhanced edge branch provides robust contour information for all targets. Second, the DSA module integrated into the spatial branch dynamically adjusts its receptive field, allowing it to generate strong and accurate activations for distant, small-scale grape clusters, mirroring its focus on nearby targets. This capability enables EFSegNet to generate more precise and robust feature representations than the baseline, consequently leading to the significant performance improvements quantified in the metrics.

3.2. Dynamic elimination performance

To evaluate the effectiveness of dynamic feature point removal, the semantic mask corresponding to each input image is integrated into the system. During the frame creation stage, dynamic feature points located within red pedestrian regions are removed based on the semantic mask, thereby reducing the negative impact of moving objects on subsequent tracking and localization accuracy. Three representative frames from the dataset constructed in this study are selected to demonstrate the effect of semantic mask-based dynamic point removal. Fig. 13(a) shows the original input image frames. Fig. 13(b) displays the feature points extracted by the original ORB-SLAM3 system. As no dynamic point filtering is applied, feature points are also extracted from moving pedestrian

regions, which degrades localization accuracy. Fig. 13(c) shows the removal results using masks generated by the original PIDNet network. Most dynamic points are effectively removed; however, due to the complex background of the vineyard and pedestrian motion, the segmentation task becomes more challenging, resulting in incomplete removal of edge features associated with dynamic objects. Fig. 13(d) presents the results using masks generated by the proposed EFSegNet network. The system successfully removes dynamic feature points both within pedestrian regions and along their boundaries, significantly reducing the influence of moving objects on SLAM performance and improving the reliability of subsequent pose estimation and map construction. The effectiveness of dynamic feature point elimination is demonstrated through comparative analysis of different approaches, as illustrated in Fig. 13. The results show that EFSegNet achieves superior performance in removing both interior and boundary dynamic feature points compared to the original PIDNet and baseline ORB-SLAM3.

The TUM RGB-D dataset [36], provided by the Technical University of Munich, contains RGB images, depth maps, and corresponding camera poses captured by a Kinect sensor. This dataset is specifically designed for evaluating and benchmarking VSLAM algorithms. In our experiments, four highly dynamic sequences and one low dynamic sequence were selected from the TUM dataset. The fr3-walking sequences represent highly dynamic scenarios in which two people move freely within an office environment, while fr3-sitting represents a low dynamic sequence where two people sit at a desk with only hand movements. The Absolute Trajectory Error (ATE) directly computes the difference between the ground truth and the estimated camera poses, providing an intuitive measure of both the accuracy and global consistency of the estimated trajectory. The Root Mean Square Error (RMSE) of the ATE compares the estimated trajectory against the ground truth trajectory; a smaller RMSE indicates that the estimated trajectory is closer to the true path.

To evaluate the localization accuracy and robustness of the proposed system, comparative experiments were conducted against the original ORB-SLAM2 and several representative dynamic SLAM algorithms, including DynaSLAM [37], DS-SLAM [38], and Panoptic-SLAM [39]. The RMSE results for all methods across different sequences are summarized in Table 4. As shown in Table 4, our system achieved the lowest RMSE in all sequences. In the fr3-walking-rpy sequence, the system

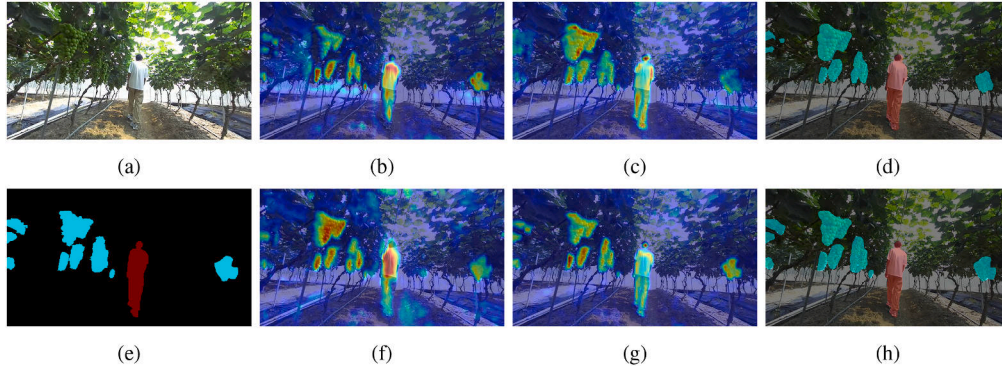


Fig. 12. Comparison of segmentation results between PIDNet and the proposed EFSegNet in the vertical sequence. (a) Image. (b) PIDNet edge branch output. (c) PIDNet Bag layer output. (d) PIDNet segmentation result. (e) Ground truth. (f) EFSegNet edge branch output. (g) EFSegNet Bag layer output. (h) EFSegNet segmentation result.

Table 4

Comparison of the RMSE of ATE in TUM sequences.

Sequence	fr3-sitting-static	fr3-walking-halosphere	fr3-walking-rpy	fr3-walking-static	fr3-walking-xyz
ORB-SLAM3	0.011	0.370	0.658	0.097	0.393
DynaSLAM	0.0075	0.035	0.162	0.043	0.019
DS-SLAM	0.0084	0.032	0.385	0.058	0.022
Panoptic-SLAM	0.0065	0.033	0.131	0.039	0.016
Proposed System	0.0052	0.031	0.082	0.037	0.015

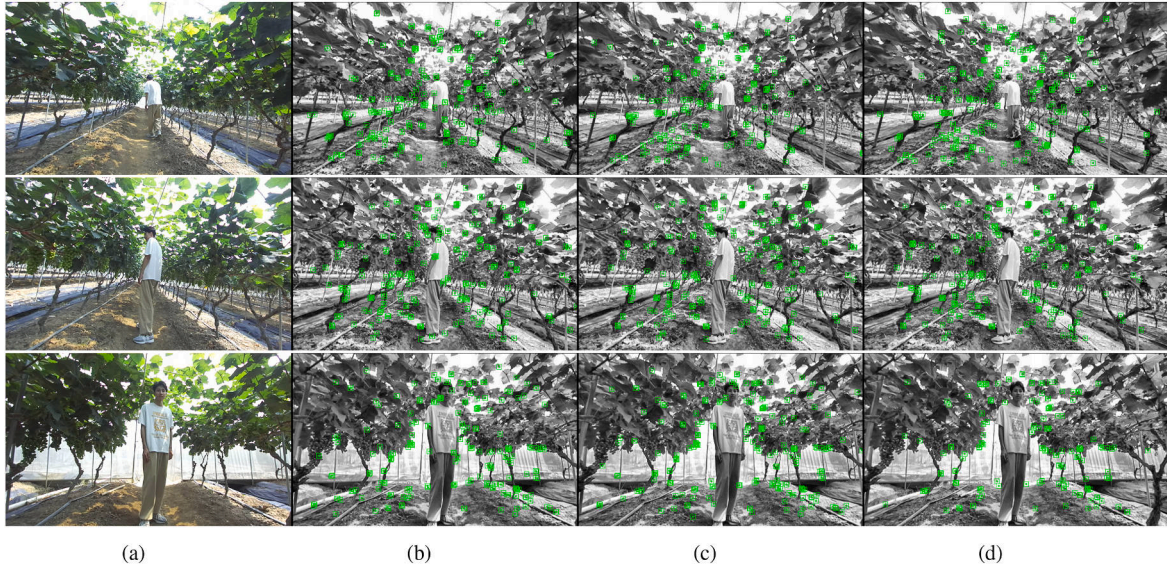


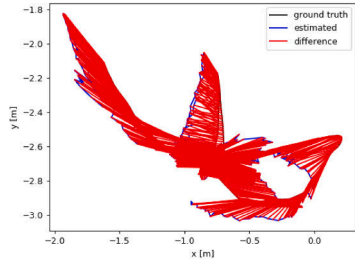
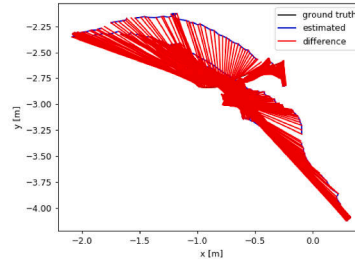
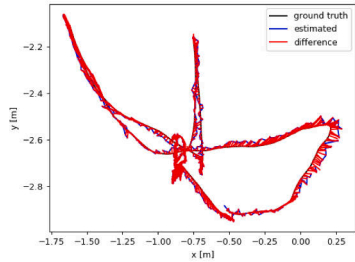
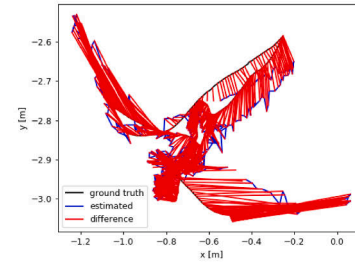
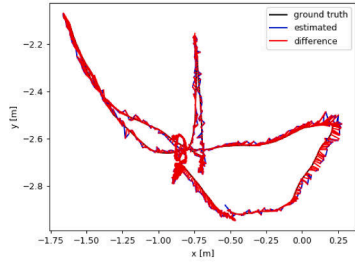
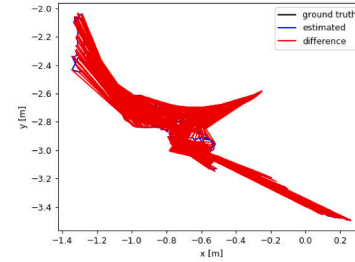
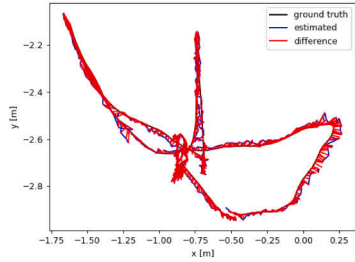
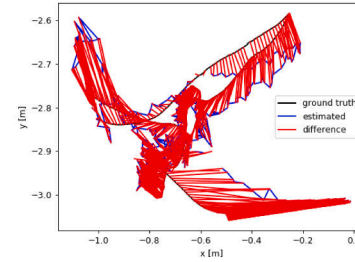
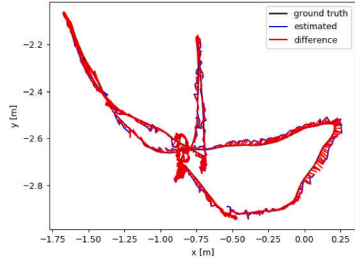
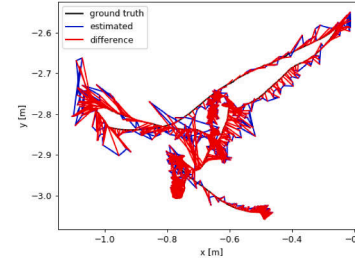
Fig. 13. Comparison of dynamic feature point elimination effectiveness across different approaches. (a) Original RGB frames showing vineyard scenes with moving pedestrian. (b) Feature points extracted by original ORB-SLAM3 without dynamic filtering. (c) Feature points after dynamic elimination using original PIDNet mask. (d) Feature points after dynamic elimination using EFSegNet mask.

attained an RMSE of 0.082, representing an 87.5% improvement over ORB-SLAM3, and 49.4% and 37.4% improvements over DynaSLAM and DS-SLAM, respectively. In the low dynamic fr3-sitting-static sequence, where dynamic motion is limited, the improvement is relatively smaller, achieving a 52.7% reduction in RMSE compared with ORB-SLAM3 and 30.7% and 38.1% reductions compared with DynaSLAM and DS-SLAM, respectively. Table 5 provides an intuitive visualization of the ATE for different algorithms on the fr3/walking-halosphere and fr3/walking-rpy sequences. The red lines represent the deviation between the estimated and ground truth trajectories. As illustrated in the figure, the developed solution demonstrates superior localization accuracy and robustness in dynamic environments compared to the other methods.

3.3. 3D Reconstruction in dynamic vineyards

To validate the 3D mapping capabilities of the proposed system in a dynamic orchard environment, we conduct a series of experiments using our built vineyard dataset. The experiments included scenes with both single-row and double-row. Data was collected from both a lateral scanning perspective and a forward-moving perspective along the rows to simulate the typical movement paths and visual changes experienced by an agricultural robot during orchard operations. Fig. 14 illustrates the mapping of a single-row trellis. Without semantic information (Fig. 14a), the system only produced a structural reconstruction. By incorporating EFSegNet, the system accurately em-

Table 5
Ground truth and trajectories estimated in the TUM sequences.

Sequence	fr3-walking-halfsphere	fr3-walking-rpy
ORB-SLAM3		
DynaSLAM		
DS-SLAM		
Panoptic-SLAM		
Proposed System		

beds fruit semantics into the map, making the grape bunches clearly discernible (Fig. 14b). This capability is further demonstrated on a more complex double-row trellis structure from both a global perspective (Fig. 15) and the robot's forward-moving perspective (Fig. 16). The system's robustness in dynamic scenes is highlighted in Fig. 17.

Without dynamic object handling, the map is corrupted by artifacts from the moving pedestrian (Fig. 17a and b). In contrast, our EFSegNet-assisted approach removes these dynamic points, resulting in a clean and coherent map where the integrity of the static structure is preserved (Fig. 17).

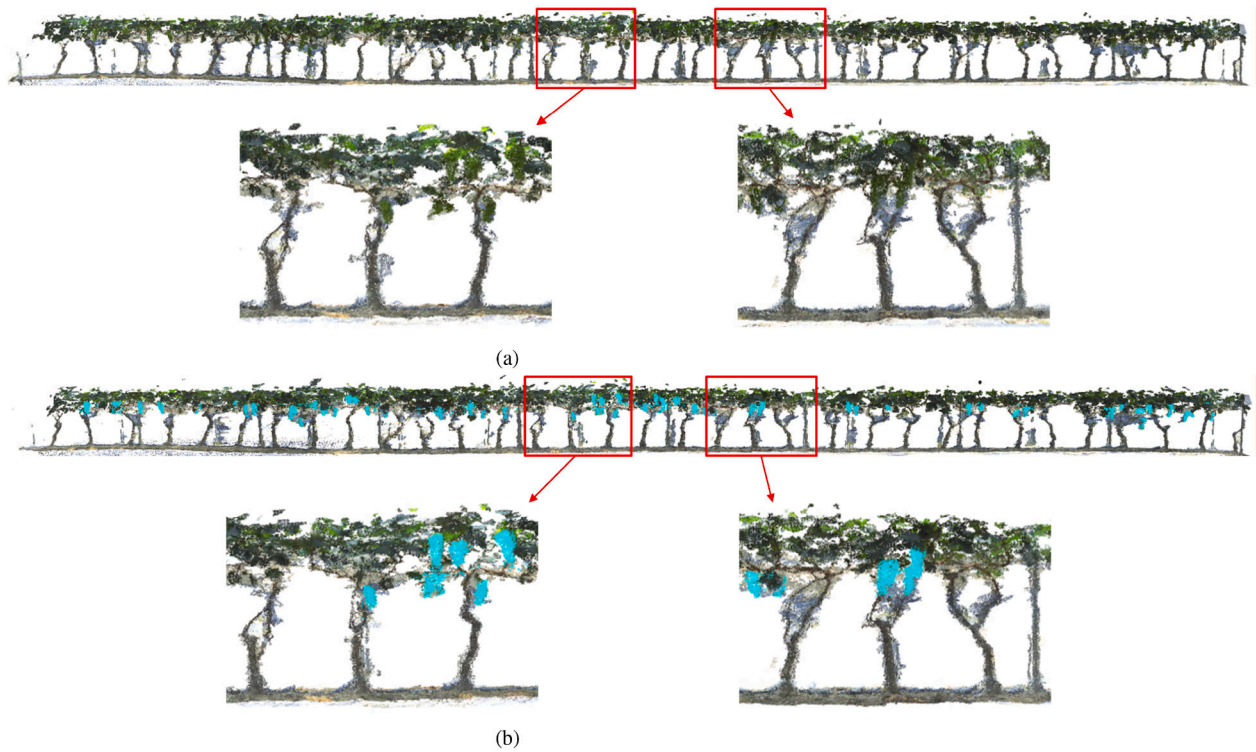


Fig. 14. 3D reconstruction comparison of a single-row trellis from a lateral scanning perspective.(a) Original 3D point cloud reconstruction. (b) Semantic 3D reconstruction with fruit regions highlighted in cyan.



Fig. 15. 3D reconstruction comparison of a double-row trellis from a global perspective.(a) Original 3D point cloud reconstruction. (b) Semantic 3D reconstruction with fruit regions highlighted in cyan.

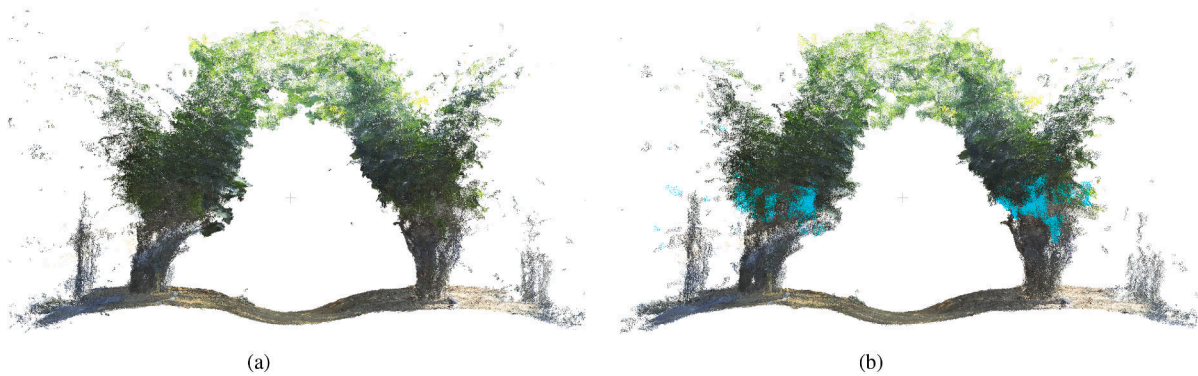


Fig. 16. 3D reconstruction comparison of a double-row trellis from a forward-moving perspective.(a) Original 3D point cloud reconstruction. (b) Semantic 3D reconstruction with fruit regions highlighted in cyan.

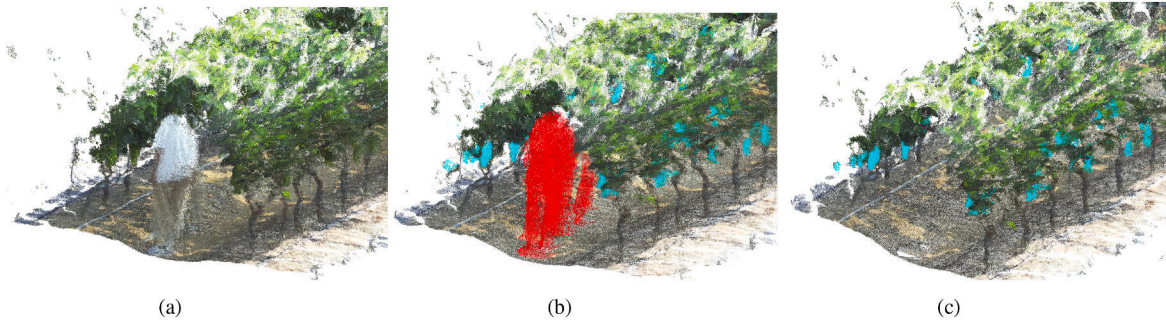


Fig. 17. Dynamic 3D reconstruction results. (a) Original 3D point cloud reconstruction. (b) 3D reconstruction with pedestrian point clouds annotated using EFSegNet semantic masks. (c) 3D reconstruction result after pedestrian point cloud removal.

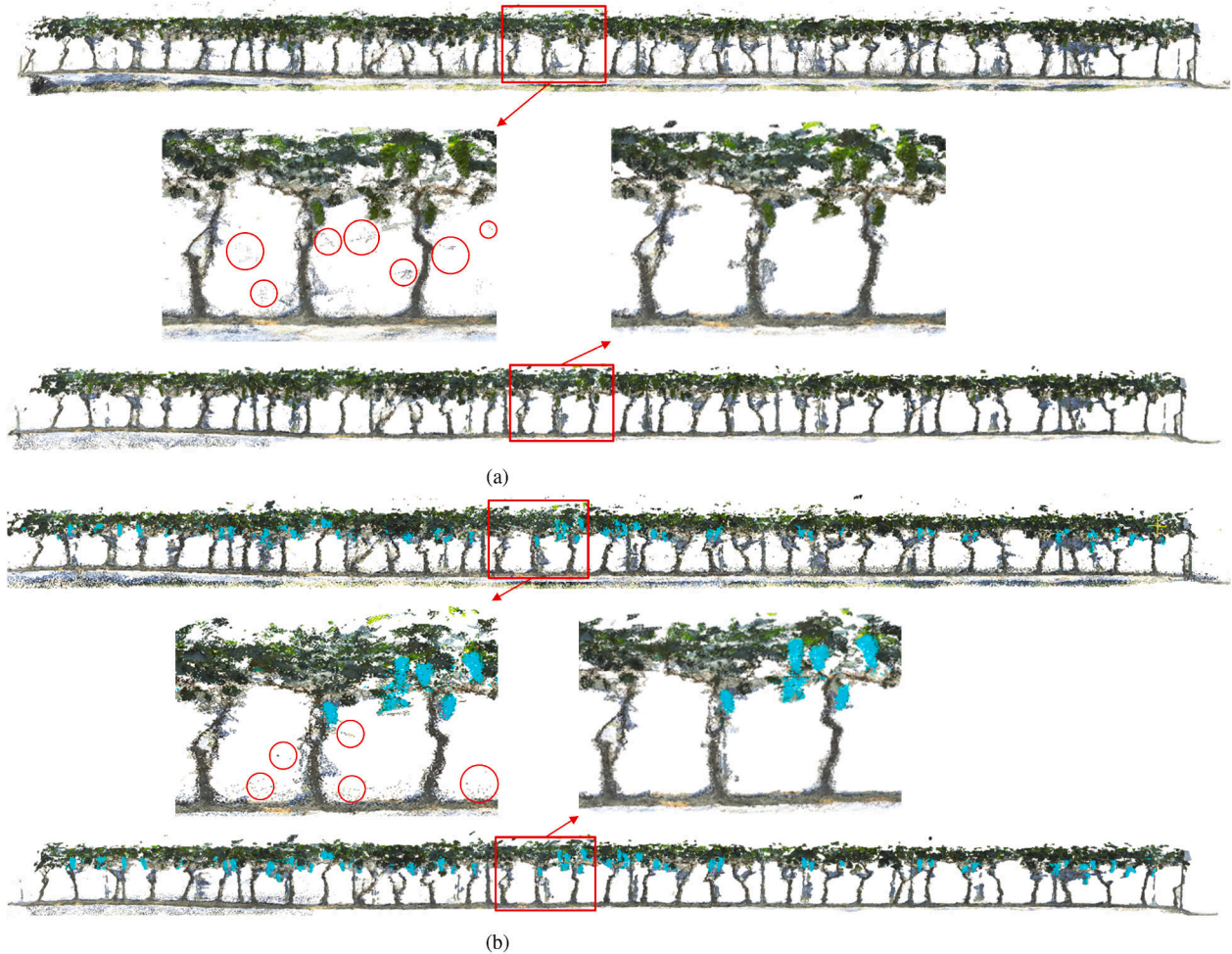


Fig. 18. 3D reconstruction of a single-row trellis from a lateral scanning perspective. (a) Comparison of point cloud map before and after filtering. (b) Comparison of semantic point cloud before and after filtering.

3.4. Map optimization analysis

Although the system achieves clear point cloud reconstruction and fruit semantic enhancement in various orchard perspectives and scenarios, a detailed analysis of local regions reveals some limitations. The resulting point cloud maps still contain a degree of noise, including outliers, non-uniform density, and fragmented structural edges. These abnormalities are mainly caused by light effects, branch and leaf shading, and depth estimation errors in the orchard environment, which may have an impact on the subsequent fruit clustering and spatial modeling accuracy if not treated.

Fig. 18 demonstrates the comparative effect of the reconstruction results before and after point cloud map and semantic point cloud map filtering optimization in a single-row grapevine scene. As shown in Fig. 18(a), there are obvious edge burrs, outlier noise and structural repetitive noise in the original point cloud map, especially in the regions with dense branches and leaves or occluded viewpoints, where the noise density is high. After the three-stage point cloud filtering, the overall point cloud map converges significantly, the local contours are clearer, the outliers are effectively eliminated, and the boundary is more regular. Further observing the semantic point cloud in Fig. 18(b), the unfiltered result shows that there are breakage or diffusion problems in

the fruit region, and semantic overlapping occurs in the junction region of the tree trunk and the fruit. After filtering, the semantic point cloud more closely matches the real fruit distribution, the semantic structure is complete, the redundant points are obviously compressed, and the consistency between the semantic information and the geometric structure of the point cloud is improved.

3.5. Fruit clustering and localization

To quantify the quality of 3D reconstruction in the vineyard, we conducted quantitative experiments with grape bunches for fruit localization accuracy experiments and bunch counting experiments.

3.5.1. Fruit localization accuracy

To obtain high-precision 3D ground truth positions of the fruit, a Robosense Helios 32 LiDAR was integrated into the experimental platform. This LiDAR provides high spatial resolution and ranging accuracy (typically < 1 cm), enabling it to precisely capture the true geometry of grape bunches and the surrounding environment. During data acquisition, the robotic platform moved slowly through the trellis rows to ensure complete scanning of each grape bunch. Subsequently, the ground truth data was generated through a meticulous manual annotation process on the collected LiDAR point clouds. Each grape bunch was manually annotated in CloudCompare by selecting its corresponding point cloud region. The geometric centroid of each annotated cluster was computed and recorded as the ground truth 3D position. To ensure rigorous spatial alignment, we perform LiDAR–stereo registration via offline LiDAR–camera extrinsic calibration using a checkerboard target, following Fu et al.’s two-stage framework (point-to-plane initialization and differentiable refinement [40]). We validated the calibration stability by performing a stereo baseline consistency check across 10 independent trials. The refined alignment achieved a mean translation residual of 2.91 mm ($\sigma = 0.51$ mm). For the ground-truth generation, the annotated LiDAR centroids were transformed into the global SLAM frame using the unified pose chain:

$$T_{WL}(t) = T_{WC}(t)T_{CL} \quad (29)$$

where $T_{WC}(t)$ represents the camera pose estimated by ORB-SLAM3.

We calculate the error between the ellipsoidal centers of fruit clusters estimated by the system and the ground truth geometric centers of the fruits. The error components are projected onto the three coordinate axes to evaluate the discrepancies along each direction, with the RMSE employed as the evaluation metric:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (value_i - \hat{value}_i)^2} \quad (30)$$

where n denotes the number of grape bunches, $value_i$ represents the ground truth value of the i -th grape bunch, and $\hat{value}_i$ denotes the predicted value of the i -th grape bunch.

For a comprehensive evaluation of the system’s localization capabilities, data was acquired under three distinct experimental sequences: (1) lateral scanning perspective, simulating cross-row observation; (2) forward-moving perspective, representing typical in-row navigation; and (3) forward-moving perspective with pedestrian interference, designed to assess robustness in dynamic scenes. The localization error metrics for each of these sequences are detailed in Table 6.

As shown in Table 6, the lateral scanning perspective (Sequence 1) yields the lowest 3D RMSE of 19.02 mm, primarily due to the sensor’s optimal distance and viewing angle, which allows for clear grape bunches representation and uniform point cloud distribution, thus facilitating precise cluster centroid extraction. Conversely, the forward-moving perspective (Sequence 2) shows a slightly higher 3D RMSE of 20.98 mm. This increase is attributed to view angle variations and localized occlusions inherent in longitudinal movement, leading to minor depth estimation drifts. Despite this, the error remains well within

Table 6

Localization error.

Sequence	RMSE (mm)				Standard Deviation (mm)
	3D RMSE	ΔX	ΔY	ΔZ	
1	19.02	10.3	11.5	11.1	4.9
2	20.98	12.9	11.7	11.7	5.4
3	23.07	13.1	15.4	14.7	6.5
average	21.02	12.1	12.8	12.7	5.6

Table 7

Experimental results of bunch counting.

Sample	True value	Predicted value	Relative error (%)
Single-row	88	84	5.7
Double-row	146	137	6.2

the acceptable range. As expected, the forward-moving perspective with pedestrian interference (Sequence 3) records the highest 3D RMSE of 23.07 mm, clearly demonstrating the impact of dynamic elements on localization accuracy. From the 3D reconstruction results of grapes in the three experimental sequences, the overall localization error of the system is well controlled. Sequence 1 exhibited the lowest standard deviation with an average value of 4.9 mm, while Sequence 2 showed a slightly higher average error of 5.4 mm due to the change of view angle. As illustrated in Fig. 19(a) for Sequence 3, the RMSE distribution with respect to distance shows that, throughout the robot’s movement, most RMSE values remain within 30 mm, with only two outlier points exceeding 40 mm. It can be observed from Fig. 19(b) that, except for a single point with slightly larger error in the Sequence 1 and 2, the rest of the data are tightly distributed and most of them are well controlled.

3.5.2. Grape bunches counting

This paper introduces a density-based clustering method called DBSCAN, which is combined with a semantic segmentation mask to automatically identify and separate spatially independent regions of grapes from a 3D point cloud. After spatial clustering is complete, the PCA method is adopted to model the point cloud shape of each cluster and extract its geometric structure. The final result of this process is shown in Fig. 20, where each ellipsoid represents a bunch of grapes, and the relative error between the actual and predicted numbers of fruits measures the accuracy of the 3D semantic reconstruction results, as shown in Eq. (31), where δ denotes the relative error, N_t the actual number of fruits and N_p the predicted number of fruits. The combined statistics of these evaluations are shown in Table 7.

$$\delta = \frac{|N_p - N_t|}{N_t} \times 100\% \quad (31)$$

The experimental results are summarized in Table 7. In the single-row scenario, the system predicted 84 bunches against a true value of 88, resulting in a relative error of 5.7%. In the more complex double-row scenario, it predicted 137 bunches against a true value of 146, with a relative error of 6.2%. These results demonstrate that the proposed method can achieve reliable and accurate fruit bunch counting in different orchard configurations. These accurate bunch count estimates can be directly used to derive per-row and per-block yield estimates in precision viticulture workflows.

Considering that fixed clustering parameters may not adapt to variations in fruit cluster density across different growth cycles, a parameter sensitivity analysis was conducted for the DBSCAN algorithm. The counting relative error was evaluated under a combination of neighborhood radius ϵ values ranging from 0.04 m to 0.06 m and minimum points MinPts ranging from 40 to 60. As shown in Table 8, the counting error remained relatively stable and ranged from 5.95% to 7.86% within the tested parameter space. The lowest error of 5.95% was observed when the parameters were set to the default values of $\epsilon = 0.05$ and MinPts = 50. A slight increase in error to 7.86% occurred with ϵ at

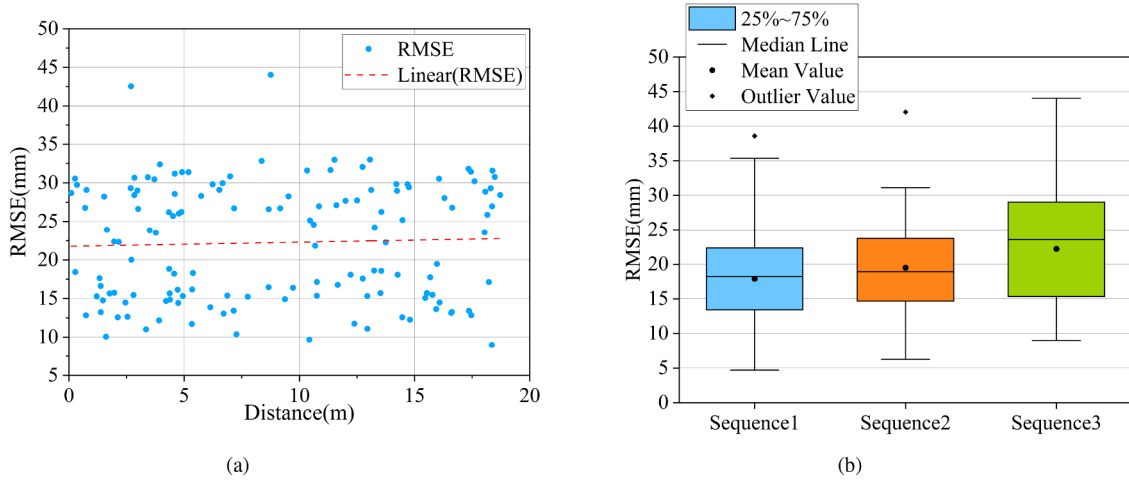


Fig. 19. Fruit localization accuracy (RMSE) results. (a) RMSE distribution by distance for the forward-moving perspective with pedestrian interference (Sequence 3). (b) RMSE distribution across different experimental sequences.

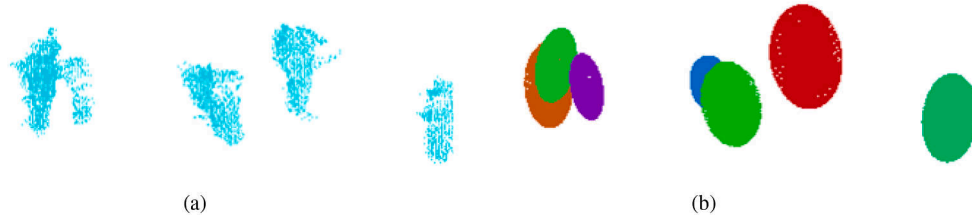


Fig. 20. Grape bunch clustering and ellipsoid modeling results. (a) Extracted grape bunches point cloud after DBSCAN clustering. (b) PCA-based ellipsoid fitting results.

Table 8

Sensitivity analysis of DBSCAN parameters on counting error.

	MinPts = 40	MinPts = 50	MinPts = 60
$\epsilon = 0.04$	6.26%	6.49%	6.72%
$\epsilon = 0.05$	6.27%	5.95%	6.04%
$\epsilon = 0.06$	7.86%	6.50%	6.16%

0.06 and MinPts at 40, which is likely due to the merging of adjacent clusters caused by the relaxed density constraints. Overall, the results indicate that the system is not overly sensitive to parameter fluctuations of approximately $\pm 20\%$, suggesting good generalization capability for practical deployment.

3.5.3. Robustness to illumination variations

Agricultural environments are subject to drastic lighting changes which can significantly affect the performance of visual sensors. To evaluate system robustness, comparative experiments were performed under three typical lighting intensities including normal light, shadows, and backlight conditions, as illustrated in Fig. 21. The quantitative impact of these variations on fruit localization and counting is detailed in Tables 9 and 10. Under normal lighting, the system achieved the lowest 3D RMSE of 19.73 mm and a relative counting error of 5.4%. In shadowed conditions, the localization error increased slightly to 21.99 mm while the counting error was 5.6%. However, backlight scenarios caused the localization error to rise to 25.22 mm with a standard deviation of 7.6 mm, and the counting error reached 7.3%. This degradation is primarily attributed to the instability of stereo matching and increased depth noise caused by local overexposure and poor texture contrast in backlight scenarios. Despite this increase, the overall performance remains within a reliable range for yield estimation tasks.

Table 9

Localization accuracy on different lighting intensities.

Condition	RMSE (mm)				Standard Deviation (mm)
	3D RMSE	ΔX	ΔY	ΔZ	
Normal	19.73	12.4	10.6	11.1	5.1
Shadows	21.99	11.5	13.8	12.7	5.8
Backlight	25.22	13.3	15.4	14.9	7.6

Table 10

Experimental results of bunch counting on different lighting intensities.

Sample	True value	Predicted value	Relative error (%)
Normal	92	87	5.4
Shadows	89	82	5.6
Backlight	96	89	7.3

3.5.4. Robustness to occlusion

To validate the necessity of the GFFM and DSA modules in complex scenarios, Fig. 22 first presents a qualitative visual comparison. As observed in Fig. 22(c), the baseline PIDNet struggles to distinguish adjacent grape bunches that are heavily overlapped and tends to merge them into single connected regions. In contrast, EFSegNet successfully delineates the boundaries of individual fruit clusters as shown in Fig. 22(d). Following this qualitative assessment, an independent quantitative analysis was performed on a heavily occluded test subset, as summarized in Table 11. While both models experience performance degradation compared to the full dataset, EFSegNet demonstrates superior robustness by achieving an mIoU of 87.62%, which outperforms PIDNet by 3.48%. Furthermore, this advantage in segmentation translates to the downstream counting task, where EFSegNet reduces the counting error to 6.23% compared to 9.71% for PIDNet. These results indicate that

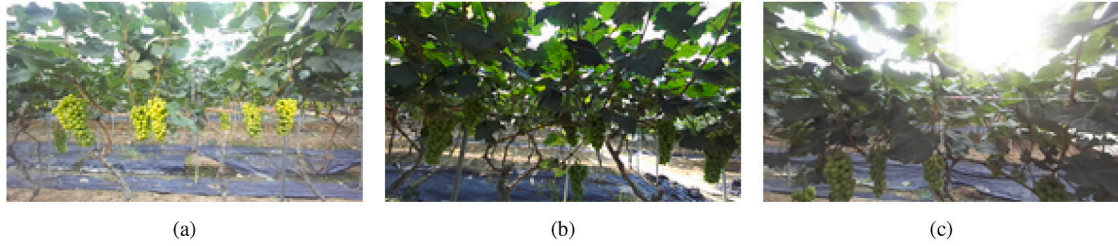


Fig. 21. Test set on different lighting intensities. (a) Normal. (b) Shadows. (c) Backlight.



Fig. 22. Visual comparison of segmentation results in a heavily occluded scenario. (a) Original image. (b) Ground truth. (c) PIDNet result. (d) EFSegNet result.

Table 11

Performance comparison on heavily occluded subset.

Method	mIoU (%)	True value	Predicted value	Relative Error (%)
PIDNet	84.14	48	43	9.71
EFSegNet	87.62	48	45	6.23

Table 12

Process time.

Module	Time (ms)
EFSegNet	7.57
Track	14.37
Point Cloud Mapping	21.4
Fruit Cluster and Localization	8.1

the enhanced edge perception effectively mitigates mis-segmentation in occluded regions, thereby directly contributing to accurate yield estimation.

3.6. System real-time performance

To evaluate the real-time processing capability of the system, the average processing time of each module of the proposed system was counted, as shown in Table 12. The system's real-time performance is primarily determined by the processing times of the semantic segmentation, track, and point cloud mapping modules. Specifically, the track module responsible for feature extraction and matching operates at an average of 14.37 ms, while the semantic segmentation module, EFSegNet, processes at 7.57 ms on average. The point cloud mapping module takes 21.4 ms per frame. In contrast, the fruit clustering and localization module operates in parallel threads with an average processing time of 8.1 ms, and thus has a limited impact on overall real-time performance. The parallel execution design ensures that all key modules remain within acceptable time bounds, supporting efficient real-time operation, and the system achieves a real-time performance of 23.1 FPS.

4. Discussion

This study introduces a comprehensive real-time 3D mapping and fruit localization system specifically designed for dynamic vineyard yield estimation. Experimental results demonstrate the system's robustness in addressing dynamic disturbances and the unstructured conditions typical of orchard environments.

4.1. Comparative analysis

Previous orchard SLAM studies, such as ORB-Livox[26], SLAM-PYE[41], and SCTNet-B & ORB-SLAM2[42], typically evaluated their systems in controlled, static scenarios, as shown in Table 13. In contrast, our system integrates dynamic object filtering through semantic segmentation, effectively mitigating dynamic disturbances encountered in realistic vineyard environments, such as moving persons. Regarding real-time performance, the system achieves a throughput of 23.1 FPS while providing 3D mapping and fruit localization. Other systems report different frame rates—for example, ORB-Livox (> 30 FPS), SLAM-PYE (23 FPS), and SCTNet-B & ORB-SLAM2 (28.36 FPS)—yet SCTNet-B & ORB-SLAM2 lacks fruit localization, and SLAM-PYE omits 3D mapping. In terms of localization accuracy, our system achieves a mean RMSE of 21.02 mm under dynamic outdoor conditions, substantially outperforming prior methods such as ORB-Livox (26 mm RMSE) and SLAM-PYE (46.3 mm horizontal RMSE). In addition to SLAM-based systems, yield estimation is also widely studied using pure-vision detection-tracking-counting (DTC) pipelines. [43] reported grape-cluster counting with pruned YOLOv5s and SORT, achieving 84.9% accuracy at 43.1–50.4 FPS. [44] reported counting performance with an R^2 of 0.85 and RMSE of 1.57, and [45] reported fruit-number prediction with an R^2 of 0.945 and yield-estimation performance of $R^2 = 0.902$. DTC relies on 2D association for de-duplication and can fail under revisits. Our SLAM world-frame anchoring reduces repeated counts and provides centimeter-level 3D localization, achieving 21.02 mm RMSE at 23.1 FPS. This is cost-effective when yield estimation requires both counting and spatial localization.

4.2. Limitations

Despite its demonstrated robustness, the proposed system faces specific limitations that affect its overall performance.

Firstly, system accuracy heavily depends on depth estimation from binocular stereo vision. While semantic segmentation helps mitigate the negative impact of dynamic disturbances, it does not resolve inherent limitations of stereo matching algorithms in regions with weak texture or poor illumination. Such conditions increase depth measurement uncertainty, resulting in gaps, structural distortions, and reduced localization accuracy, especially during longitudinal vineyard traversal.

Secondly, while the three-stage filtering strategy effectively removes ghost points and noise, it inevitably incurs a loss of geometric details. The voxelization process, which substitutes point clusters with centroids, coupled with aggressive outlier removal, tends to over-smooth

Table 13
Comparison of yield estimation systems: SLAM baselines vs. Pure Vision (DTC) methods.

System	ORB-Livox [26]	SLAM-PYE [41]	SCTNet-B & ORB-SLAM2 [42]	[43] (DTC)	[44] (DTC)	[45] (DTC)	Proposed
Object	Apple	Playa	Eggplant	Grape clusters	Apple (orchard)	Camellia oleifera fruit	Grape
Sensor	LiDAR, Stereo Camera	GNSS, Binocular Camera, IMU	Depth Camera, LiDAR, RTK	RGB Video Camera	RGB Camera (image + video)	RGB Camera	Stereo Camera
Hardware	Intel i5-8300H GTX-1060	NVIDIA RTX-3060	Intel Xeon w5-3423 NVIDIA RTX A5000	NVIDIA RTX 3090	AMD EPYC 9754S NVIDIA RTX 4090(24GB)	AMD EPYC 9754S NVIDIA RTX 4090(24GB)	Intel i5-12500H NVIDIA RTX-3060
Scenario	6 (outdoor)	F1 (greenhouse)	Greenhouse	Grape Field	7311 imgs & 24s video	Tree-level yield estimation	3(Dynamic Orchard)
RMSE(mm)	26	46.3 (Horizontal) 54.0 (Vertical)	Without fruit localization	-	-	-	21.02
SD(mm)	13.6	-	-	-	-	-	5.6
Metric	-	-	-	-	-	-	-
Real-time(FPS)	> 30	-	28.36	Counting accuracy 84.9% Up to 50.4	Counting: $R^2 = 0.85$ RMSE = 1.57 Not reported	Fruit $R^2 = 0.94$ Yield $R^2 = 0.90$ Not reported	Relative error 5.7% / 6.2% (Avg 5.95%) 23.1
Dynamic filtering	x	x	x	x	x	x	✓

high-frequency features. Consequently, sparse but valid structures, such as thin tendrils or fruit peduncles, may be inadvertently filtered out. Although this does not impact the volume-based counting of grape bunches, it limits the potential extension of the current system to more delicate phenotyping tasks that require sub-centimeter structural fidelity.

Thirdly, the current strategy for handling dynamics focuses on rigid objects such as pedestrians and does not incorporate specific mechanisms to handle environmental dynamism involving deformable elements, specifically foliage swaying caused by wind. Although the back-end optimization and filtering modules effectively suppress these oscillatory motions of small magnitude as point cloud noise, the lack of a dedicated mechanism for tracking deformable objects means the system may face stability challenges. This limitation becomes particularly relevant under conditions where environmental factors induce non-rigid deformations that exceed the capacity of robust adjustment.

5. Conclusion

This study presents a real-time semantic 3D mapping and fruit localization system designed to address the challenges of yield estimation in dynamic and unstructured vineyard environments. The proposed system integrates the EFSegNet, enhanced with Gradient Feature Fusion and Deformable Spatial Attention, into the ORB-SLAM3 framework to effectively eliminate dynamic feature points while ensuring high-quality semantic mapping. Experimental results demonstrate that EFSegNet achieves an mIoU of 89.45% and a Boundary F-measure of 82.78%, significantly refining fruit boundaries and separating overlapping clusters. This segmentation superiority translates into robust downstream performance, maintaining a low counting error of 6.23% even in heavily occluded scenarios and showing stability under varying illumination conditions. Overall, the system realized an average 3D localization accuracy of 21.02 mm with a processing speed of 23.1 FPS, confirming its effectiveness for autonomous orchard inspection.

Future work will focus on three key directions to further improve system adaptability. First, we plan to introduce multi-sensor fusion strategies, incorporating LiDAR or IMU data, to compensate for the inherent limitations of visual depth estimation in low-texture or strong backlight regions. Second, we aim to develop adaptive filtering and clustering algorithms that automatically adjust parameters according to local point cloud density, enhancing generalizability across different fruit varieties. Finally, future research will address environmental dynamism involving non-rigid deformations, specifically modeling foliage swaying caused by wind, to further ensure tracking stability in complex field conditions.

CRedit authorship contribution statement

Runbin Yang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Data curation; **Xilong Zhang:** Writing – review & editing, Writing – original draft, Validation, Methodology; **Yanyang Chen:** Writing – review & editing, Visualization, Data curation; **Xiaotao Huang:** Writing – review & editing; **Sang Feng:** Resources, Project administration, Conceptualization.

Data availability

I have shared the dataset associated with this manuscript at the following public repository: <https://github.com/yangrb2333/grl-slam/tree/main>

Ethical statement

This paper adheres to the fundamental principles of academic integrity and relevant ethical norms. The content is original, free from plagiarism, fabrication, or inappropriate data manipulation, with all sources fully and accurately cited. No potential conflicts of interest exist.

among the authors. All contributors confirm compliance with academic ethical standards and take full responsibility for the accuracy and integrity of the work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.atech.2026.101784](https://doi.org/10.1016/j.atech.2026.101784)

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